

Thin Detectors for the CHICSi ΔE -E Telescope

L. Evensen¹, *Member, IEEE*, T. Westgaard¹, *Member, IEEE*, V. Avdeichikov², L. Carlén²,
B. Jakobsson², Y. Murin², J. Mårtensson², A. Oskarsson², A. Siwek², H.J. Whitlow³,
E.J. van Veldhuizen⁴, L. Westerberg⁴, and M. Guttormsen⁵

¹SINTEF Electronics and Cybernetics, P.O.Box 124 Blindern, N-0314 Oslo, Norway

²University of Lund, Div. of Cosmic and Subatomic Physics, P.O.Box 118, S-221 00 Lund, Sweden

³Lund Institute of Technology, Dept. of Nuclear Physics, P.O.Box 118, S-221 00 Lund, Sweden

⁴Uppsala University, Dept. of Radiation Sciences, P.O.Box 535, S-751 21 Uppsala, Sweden

⁵University of Oslo, Dept. of Physics, P.O.Box 1048 Blindern, N-0316, Oslo, Norway

Abstract

A pilot series of 10 μm to 15 μm thin silicon detectors has been made for the ΔE -E telescopes in the CHICSi detector system. This system will operate at the CELSIUS heavy ion storage ring in Uppsala, Sweden. ΔE -E telescopes provide isotope identification and energy determination of fragments from nuclear collisions. The thin detectors are made as p-i-n diodes in thin etched membranes in 280 μm thick silicon wafers. The membranes are made with anisotropic etching using 25 w% tetramethylammonium hydroxide (TMAH) solution. The etch speed of this solution is very uniform across a wafer. As a result detectors with uniform thickness can be produced. The etch depth varies with less than $\pm 0.3 \mu\text{m}$ over a wafer and the surface microroughness is in the range from 2 to 4 nm. Each detector has a 10.0 mm $\times$ 10.0 mm active area on a 10.2 mm $\times$ 10.2 mm membrane surrounded by a 1.1 mm wide supporting frame. The detectors have leakage currents in the active area of approximately 0.5 nA at 20 V. The breakdown voltage of the detectors is above 100 V. Evaluation experiments with telescopes consisting of a thin detector in combination with a thick detector have shown excellent isotope separation capabilities. Mass separation of ^6Li and ^7Li is clearly observable.

I. INTRODUCTION

The CELSIUS Heavy Ion Collaboration (CHIC) has developed a telescope detector array with silicon p-i-n detectors (CHICSi) that will be used to study intermediate mass fragments (IMF's). These fragments can for example be produced in heavy ion collisions at the CELSIUS ion storage ring at The Svedberg Laboratory (TSL) in Uppsala, Sweden. Each telescope is built from three or four detector elements as shown in Figure 1. The first elements of the telescopes are detectors with silicon thickness D_{Si} of 12 μm with a 280 μm or 500 μm Si detector behind. These are operated in a ΔE -E configuration which allows charge and mass identification. Telescopes in the very forward directions have additional E-detectors made of cerium doped Gadolinium Silicon Oxide (GSO) scintillators read out by photodiodes in order to measure high energy fragments. A 280 μm Si detector (HE

flag) at the rear of the telescope is used to identify high energy fragments that pass through all detectors. The detector telescopes are assembled in rings forming a barrel shaped configuration with the detectors directed towards the interaction point. The telescope rings cover scattering angles from 15° to 160° .

A major technological challenge has been the development of technology for production and testing of the thin detectors for the ΔE -E telescopes, since 1200 detectors are required for the experiment. In a thin detector with uniform silicon thickness D_{Si} the energy loss ΔE of a fragment is a function of proton number Z and mass number A . (See e.g. [1] for details of the formalism.) This makes it possible to determine energy, proton number, and, for not too high A , the mass of an IMF with a ΔE -E detector telescope. Several parameters for the thin detectors need to be optimized to give good mass and energy resolution. In order to extend the capability to measure fragments with higher Z and low energy the absolute thickness of the ΔE -detector should be minimized to the lowest practical one. Unfortunately, thinner detectors imply a smaller signal and a higher capacitance which cause a severe reduction in signal-to-noise ratio. Another important factor is the consequence of detector thickness variations since the energy loss is proportional to the local detector thickness. The

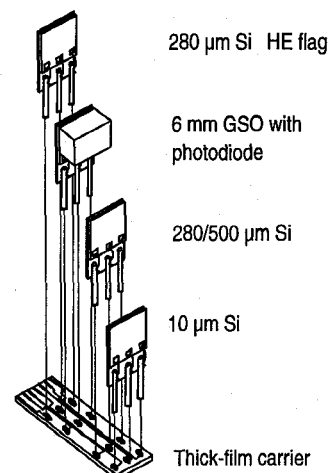


Figure 1: CHICSi detector telescope.

absolute value of thickness variations is in practice almost independent of detector thickness with the process which has been developed. As a consequence, the ratio of detector thickness variations to average thickness increases with decreasing detector thickness. Identical particles passing the detector in positions with different thickness will produce different charge signals. This effect reduces the resolution of the detector telescope. The minimum detector thickness is therefore set by the value of detector thickness variations and the electronic noise of the detector system. To achieve acceptable resolution with a 10 μm detector the thickness variations should be within 0.5 μm .

An approach to achieve thin detectors which is compatible with standard processing equipment is to process thin p-i-n diodes on wafers with etched membranes. The design and processing of the thin detectors are then similar to that of conventional silicon detectors. The focus in this study has been on improving the thickness uniformity and reducing the surface microroughness caused by the etch procedure. Tests have been carried out to ensure that the silicon detectors will be compatible with the operation requirements for experiments at the CELSIUS storage ring. The thin detectors have been characterized both mechanically and electrically and their capability for energy and mass separation has been demonstrated.

II. ETCHING OF THIN DETECTORS

Several approaches to fabrication of thin detectors have been published [2], [3], [4], [5]. An intuitively straight forward process is to use wafers of adequate thickness. The minimum, commercially available, wafer thickness is in the 10 μm range. Processing of wafers with thickness below 30 μm demands non-standard processing equipment and procedures. Detector handling and mounting are also complicated. For this reason thin detectors have usually been manufactured by etching thin areas for the active part of the detector. With this technique thicker substrates can be used, and thin membranes can be realized with standard processes for detector fabrication. The etching process has to meet three requirements to be useful for detector fabrication:

- It must be possible to etch membranes with a controlled thickness. This can be achieved by using an etch with a stable etch rate or utilizing a stop layer.
- The increase of thickness non-uniformity during the etching process has to be minimized. A common problem is grooves at the border of the etched areas.
- The etch should not increase short-range surface microroughness.

The growth of an epitaxial silicon layer is frequently exploited to achieve stopping of the etching and improve uniformity [4]. This layer can either be highly p-type doped, or n-type doped if it is used with an electro-chemical etching process. The epitaxial silicon is not ideal as detector material because of the density of impurities and defects. Both

isotropic and anisotropic etchants can be used for etching. A common isotropic etchant is a mixture of hydrofluoric acid, nitric acid and acetic acid (HNA). The reaction with silicon is extremely exothermic and the etch speed increases considerably with temperature thus making it difficult to achieve a defined and uniform thickness. The mixture is very aggressive and gold or silicon nitride must be used as a mask material. Commonly used anisotropic etchants have been ethylenediamine-pyrocatechol-water (EPW) or potassium hydroxide (KOH) solutions [6]. A problem with both of these etchants has been to achieve uniform membranes with smooth surfaces [4], [5], unless an etch stop layer is used. A process with anodic oxidation of silicon has been reported to circumvent the uniformity problem, but it introduces problems with high doping density and complexity of etching [4]. Thermally grown silicon oxide can easily be used as an etch mask in combination with EPW, while the high etch speed of silicon oxide in KOH requires very thick oxide layers.

Recently a considerable interest has emerged for etching of 3-dimensional silicon micromechanical devices with tetramethylammonium hydroxide (TMAH) because of its compatibility with CMOS processing [7]. Several technological advantages promote the use of TMAH which is known from semiconductor processing as an alternative for metal ion free developer for positive photoresist. TMAH etches silicon dioxide very slowly, almost four orders of magnitude slower than the (100) plane of silicon [7], allowing thermal oxides to be used for masking. This simplifies processing compared to KOH and one avoids the highly toxic and carcinogenic EPW solutions. However, TMAH has been reported to give pyramidal hillocks at concentrations below 15 w% [7]. The quality of the etched surface can be very good at high concentrations (above 25 w%). In section IV a surface microroughness of 3.6 nm is reported for etching with TMAH, which is considerably better than the 33 nm reported for etching with KOH [5]. The etching speed of TMAH is comparable to that of KOH [6], [7], and the etch depth uniformity is excellent without problems with groove formation along the edges of the etched area. At a controlled temperature and with a sufficient volume of etching solution the etch rate is stable. This makes the etch ideal for etching of membranes in float zone silicon with time control for obtaining the target membrane thickness. This method was exploited for producing the ΔE -detectors. The approach requires a precise thickness measurement before the final etching step.

III. DESIGN AND PROCESSING

The 12 μm , 280 μm and 500 μm thick CHICSi silicon detector elements have all been produced with a common front side mask design. The active area of the ΔE -detectors is a 10.0 mm $\times$ 10.0 mm p-n junction on a 10.2 mm $\times$ 10.2 mm membrane of n-type silicon. A cross section of the detector is shown in Figure 2. The entire active area is covered by Al which is connected to the p-type diffusion by a narrow contact

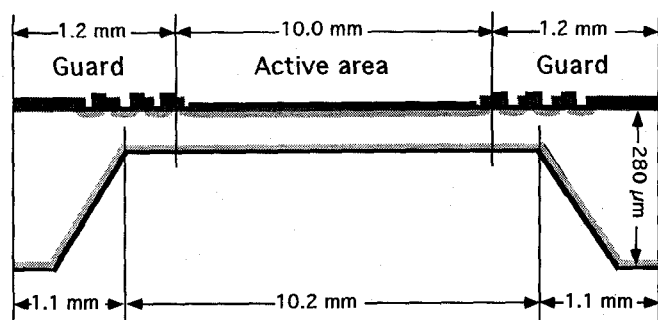


Figure 2. Schematic cross section of a ΔE -detector. Black is aluminum, dark gray is oxide, n^+ - and p^+ -diffusions are light gray.

hole along the periphery. A multiguard structure surrounds the active area. The thin region of ΔE -detectors extends out to halfway between the first and second guard rings. Only the first guard ring is of importance for these detectors as they will be operated at low or moderate voltages. This guard ring is grounded and will collect leakage current generated in the frame structure. The distance from the membrane border to the edge of the device is 1.1 mm in order to have a frame which ensures the mechanical integrity of the detectors. A narrower frame would be advantageous because the dead area of the detector system can be reduced, but due to the large number of detectors (~ 2000) for the total detector system it is important to have robust detectors. The frame is wider along the bottom edge where a part of the active area extends out on thick silicon. This design enables external connection with bond wires, but the pads are large enough to allow conventional wires to be glued to the pads.

The full guard structure was optimized for use on E-detectors processed on 500 μm thick silicon. The multiguard structure uses concentric, floating p^+ -type diffusions with the inner ring contacted (grounded) as for the ΔE -detectors. The other rings are biased by the reach-through (punch-through) mechanism. Metal field plates connected to each p^+ -type diffusion stabilize the surface potential on the oxide between the diffusions and thereby improve long term stability [8]. To ensure stable operation at full depletion and maximum charge collection over the life-time of the experiment, the detectors must also withstand considerable overdepletion at the initial testing. The width of 1.2 mm around the active area allows the use of a very stable guard structure. The stabilization of the detectors is important as the primary signal particles are highly damaging nuclear fragments which the detectors must tolerate without serious degradation of performance. It is also required that the detectors survive a low energy background of electrons and X-rays. Since the detectors will be operated in an ultrahigh vacuum environment they must withstand repeated bakeout to 125 - 150 $^{\circ}\text{C}$. No low vapor pressure materials can be used for mounting the detectors because ultrahigh vacuum conditions are required for the experiment.

High resistivity (5000 Ωcm) phosphorus doped wafers from TOPSIL were used for processing the detectors. The wafers had diameter 100 mm and thickness 280 μm , and the orientation was (100) for compatibility with the anisotropic etching. Both wafer sides were polished to SEMI specifications. Initially an 800 nm thick thermal oxide layer was grown for masking and passivation purposes. A standard photolithography process with etching of the oxide defined the oxide etch mask for etching of thin membranes on the reverse side of the wafers. The membranes were etched at 80 $^{\circ}\text{C}$ in a 25 w% solution of TMAH in water. A two step etching process with time control allowed precise control of the membrane thickness. Openings for the boron implant were defined on the front side of the wafer by lithography after careful cleaning. A thin layer of thermal oxide served as an implant window for boron on the front side and phosphorus on the back side. The implants were activated by an annealing process in N_2 at 900 $^{\circ}\text{C}$. Contact holes were defined by lithography on the front side leaving the back side unprotected. Following etching of the contact holes 0.2 μm thick films of aluminum were sputter deposited on the front and back sides. The front-side metallization was patterned before alloying at 450 $^{\circ}\text{C}$ in forming gas. The wafers were subsequently cut into detector chips with a diamond saw. To produce detectors with a narrower frame, laser cutting of wafers might be a better alternative than diamond saw cutting.

IV. EVALUATION

The wafers and detector chips have been characterized mechanically, electrically, and as detectors.

A. Mechanical Evaluation

One problem with production of thin detectors is to measure the absolute thickness and uniformity. The 10 μm range is outside the range where calibrated thickness standards for mechanical measurements are readily available. Several methods for thickness measurement can be conceived but most of them are not compatible with large volume production of thin detectors. Energy loss measurements of protons or alpha particles are excellent for uniformity measurements but are difficult to calibrate for absolute thickness measurements. Light attenuation has similar problems and must be calibrated, furthermore source and detector variations introduce too large uncertainties to be of practical use. As the index of refraction for silicon is well known, measurement of interference patterns with an FTIR instrument is considered an accurate method, but equipment for these measurements on small areas was not available at SINTEF. During etching mechanical measurements of membrane thickness were performed with a double stylus type instrument (TESA Electronic GND22 with two TESA GT10 measurement probes). Five fixed points were measured on each wafer and the average membrane thickness was used to determine the remaining etch time. The mechanical system was calibrated with shims that could be traced to Norwegian national standards. For 48 wafers a

standard deviation of 0.7 μm for the wafer average of the membrane thickness was achieved. The variation on a single wafer can be larger, because some wafers were wedge-shaped.

The thickness measurement for the final selection of finished ΔE -detectors was done by measuring the detector capacitance at a fixed voltage which was higher than the voltage needed for the depleted region to extend into the n^+ -type doped region. This method measures the width of the depletion zone of the detector while the total thickness can be computed taking the thickness of dead layers into account. For comparison with mechanical stylus measurements it is better to calculate the total detector thickness D_{diode} including oxide and metal layers in stead of the silicon detector thickness D_{Si} . The advantages of the method are that it is compatible with the electrical testing of detectors, it is non-destructive, and it is useful for detector response calibration. In contrast, testing with particle beams is time consuming and expensive. The capacitance method is accurate if an independent absolute measurement of the thickness is available so that a precise value for the dead layer thickness can be extracted. The basis for the calculation of thickness is the formula for the capacitance C of a p-i-n-diode:

$$C = \epsilon_{\text{Si}} \epsilon_0 \frac{A_{\text{active}}}{w_{\text{depl}}} \quad (1)$$

ϵ_{Si} is the relative static dielectric constant for silicon, A_{active} is the area of the pn-junction, and w_{depl} is the width of the depletion zone. For an applied voltage above the depletion voltage the total detector thickness D_{diode} is:

$$D_{\text{diode}} \approx \epsilon_{\text{Si}} \epsilon_0 \frac{A_{\text{active}}}{C} + x_n + x_p + d_{\text{oxide}} + d_{\text{Al}} \quad (2)$$

x_n and x_p are thicknesses of dead layers in silicon on each side of the detector, d_{oxide} is the front oxide thickness and d_{Al} is the total thickness of aluminum layers. The total dead-layer thickness estimated from x_n , x_p , d_{oxide} , and d_{Al} is about 1.4 μm .

If a reliable independent measurement of thickness is available it is better to use:

$$D_{\text{diode}} \approx \epsilon_{\text{Si}} \epsilon_0 \frac{A_{\text{active}}}{C} + \Delta_{\text{empirical}} \quad (3)$$

$\Delta_{\text{empirical}}$ is an empirically determined thickness correction which includes all dead layers. This dead-layer has been estimated by mechanically measuring the membrane thickness in the center of the membrane and subtracting the depletion thickness calculated from a capacitance measurement. A value of 1.1 μm for $\Delta_{\text{empirical}}$ is obtained in this way, based on measurements on 98 detectors. The discrepancy between the dead layer estimate of 1.4 μm from process parameters and 1.1 μm from mechanical measurements is within the combined measurement errors of both methods. For the first estimate it is difficult to determine x_n and x_p with a better precision than ± 0.1 μm , while the offset error in the mechanical

measurement is an important error source for the second estimate. As a consequence capacitance measurements can be used to obtain reliable values for the average detector thickness. An average membrane thickness value of 11.4 μm with a standard deviation of 0.8 μm was achieved for 630 ΔE -detectors. Each CHICSi telescope ring will be built from detectors within a small thickness range to have a uniform response in each ring.

Thickness uniformity is important for the detector quality and the yield of the production. Too large variations over a membrane will cause insufficient energy resolution. If the variations over the wafer are too large, few membranes will have the specified thickness. With a time dependent thickness definition, the thickness uniformity of the starting wafer is most critical as it will influence the number of diodes on a wafer that are within the thickness specification. In the extreme case, with an ideal etch, the wafer thickness uniformity will define the thickness uniformity across each detector. Measurements of etch depth with a Tencor Alpha-Step 500 profilometer indicate that the etch depth variation, measured from the starting surface of the wafer down to the etched membrane, varies by as little as ± 0.1 μm . This is about the level of accuracy of the instrument. Formation of grooves along the edges could not be observed for TMAH etching with the instruments available. This implies that the etching process itself does not introduce uniformity problems for the membranes. Most wafers that were measured had an etch depth variation across the wafer which was less than ± 0.3 μm . Temperature variations in the etch bath can cause such variations. This variation alone does not introduce any significant yield reduction. Thus for the TMAH etch the most important parameter determining the uniformity was the thickness uniformity of the starting wafer, because the etch process is nearly ideal with respect to uniformity. In the worst case it was observed as much as 7.4 μm thickness variation of membranes across a wafer which would lead to 1.3 μm variation across a detector. Few detectors of this wafer had the correct thickness, and none had satisfactory uniformity. This implies that the thickness uniformity of the starting wafer will determine how many devices from each wafer that are within specifications.

The short-range surface microroughness is characteristic of the etchant and can indicate the ability to etch uniformly. The surface microroughness R_a was measured with a Tencor Alpha-Step 500 profilometer along a 500 μm long path near the center of the membrane. Leveling was performed between the positions at 50 and 450 μm to reduce the influence of membrane bulging caused by the passivation oxide. R_a is the arithmetic average deviation of the absolute values of the microroughness profile from the mean value along the measurement path. A surface microroughness R_a in the range 2 - 4 nm was measured. A similar measurement was performed with a WYKO optical scanning interferometer which confirmed the mechanical measurements giving $R_a = 3.6$ nm.

Some detectors have been tested using a collimated beam of α -particles from a ^{241}Am source. A circular collimator with 1 mm diameter defined the spot on the membrane. The beam was scanned in steps over the detector. The residual energy of the α -particles after they had penetrated the thin detectors was measured with a silicon p-i-n-detector. The thickness of the membrane in the probed spots can be calculated from the energy loss. Results for 5 spots on a detector are shown in Table 1.

Table 1.

α -particle energy-loss thickness measurement on a detector. The thickness measured is the total detector thickness D_{diode} . The center of the coordinate system is the center of the active area.

Position [mm]	Thickness [μm]
(0,0)	12.73
(3.5, -3.5)	12.51
(3.5, 3.5)	12.80
(-3.5, -3.5)	12.55
(-3.5, 3.5)	12.64

B. Diode evaluation

Every diode has been electrically characterized by measuring the capacitance C and the leakage currents I_{active} and I_{guard} as a function of bias V . The depletion voltage is determined to be about 2.5 V from $1/C^2$ versus V curves. This is significantly higher than the value which is expected from the substrate resistivity, but for these thin detectors depletion of the tail

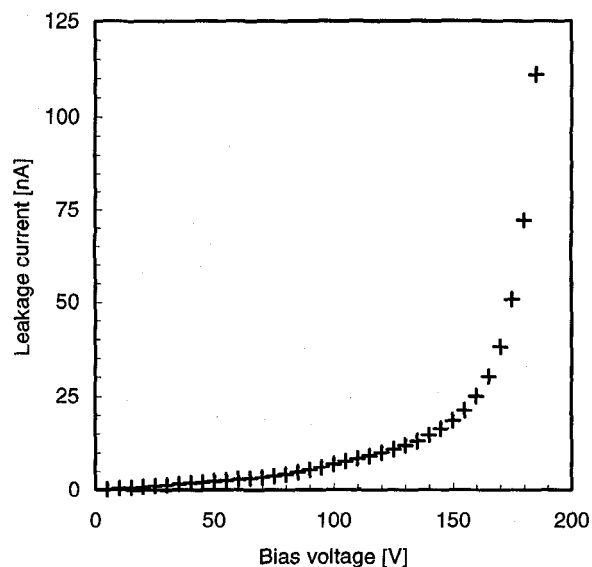


Figure 3: Leakage current as function of bias voltage for a detector with 12 μm silicon thickness.

regions in the dopant profile for n-type and p-type diffusions contribute significantly to the shape of the curve. The leakage current I_{active} in the active area is typically 0.5 nA at 20 V (approximately 8 times overdepletion) with a guard leakage current I_{guard} of typically 0.2 nA at 20 V. Some detectors have been tested with voltages up to 200 V. The leakage current increases with increasing bias voltage as shown in Figure 3. Due to the large values of the electric field in thin detectors the bulk avalanche effect will cause electrical breakdown when the bias voltage reaches a sufficient value. This breakdown voltage is influenced by the detector thickness, and it will decrease with decreasing thickness.

C. Detector performance

To provide a source of IMF's for testing of the detectors a detector telescope was set up in the CELSIUS target chamber at a backwards scattering angle of 140° . No tendency of instability in the UHV environment was observed. A beam of protons impinged on a Kr gas target. The energy of the proton beam was slowly ramped from 150 to 500 MeV. The ΔE -E telescope using 12 μm ΔE -detectors and 280 μm E-detectors resolved intermediate mass fragments with atomic numbers up to $Z = 7$. Figure 4 shows a point plot of ΔE vs. E for one of the telescopes that has been tested. The resolution is sufficient to separate ^6Li from ^7Li . Inspection of the measurement data reveals that ^7Be and ^9Be are resolved as well, but this is not evident from Figure 4 as the production of Be fragments is lower than that of Li fragments for this particular reaction. The low energy threshold which is set by the thickness of the ΔE -detector corresponds to about 1A MeV, while the high

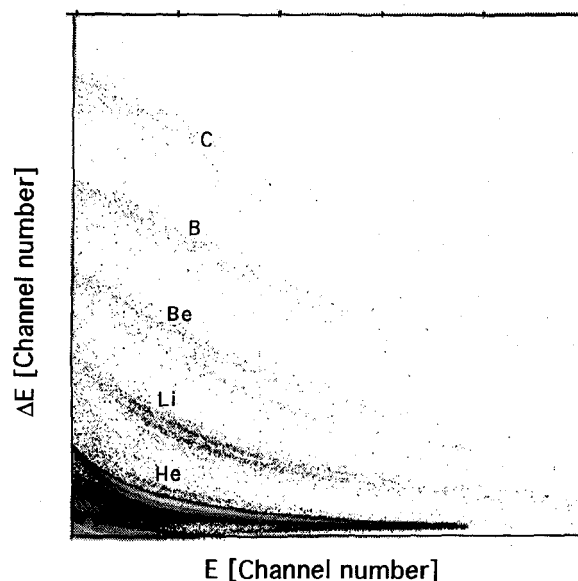


Figure 4: ΔE vs. E plot for intermediate mass fragments emitted at 140° from 150-500 MeV protons incident on a Kr gas target. Light gray indicates the highest density of fragments. The full scale readings are approximately 31 MeV for the E-axis and 23 MeV for the ΔE -axis.

energy threshold is about 6A MeV for the test telescope. This value for the high energy threshold is due to the 280 μm thickness of the E-detector. IMF's with higher values of Z have been found in evaluation experiments with heavier ions performed with CHICSi test telescopes in the CELSIUS target chamber, for instance IMF's up to $Z=14$ have been resolved for the reaction ^{14}N on ^{128}Sn . For the CHICSi experiment the upper limit in energy of IMF's will be 60A MeV in forward telescopes, 14A MeV in mid ($\sim 90^\circ$) telescopes and 6.5A MeV in backward telescopes. These regions of IMF detection are relevant both with respect to empirical knowledge and theoretical estimates [9].

V. CONCLUSIONS

Silicon detectors in the 10 μm thickness range can be produced in commercial quantities for use in telescopes for identification and energy measurement of IMF's. Etched detector membranes of sufficient quality for this purpose have been made by etching in TMAH. The diode processing yielded stable, low leakage current diodes with high voltage capability. Test telescopes have demonstrated that the detectors provide isotope separation capability for light and medium mass fragments.

For future systems it is important to note that the etching of membranes in combination with standard processing with ion implantation gives a moderate cost alternative for large volume applications.

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